

Answer all questions. Write your answers in the spaces provided.

1. A sequence $a_1, a_2, a_3, \dots$ is defined by

$$a_{n+1} = 4 - a_n$$

$$a_1 = 3$$

Find the value of

(a) (i) a_2

(ii) a_{107}

(2)

(b) $\sum_{n=1}^{200} (2a_n - 1)$

(2)

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$$(x-4)^2 \geq 2x-9 \tag{3}$$

(ii) Show that the following statement is untrue.

$$2^n + 1 \text{ is a prime number for all values of } n, n \in \mathbb{N} \quad (1)$$

- (3)

5. A company makes a particular type of watch.

The annual profit made by the company from sales of these watches is modelled by the equation

$$P = 12x - x^{\frac{3}{2}} - 120$$

where P is the annual profit measured in thousands of pounds and $\pounds x$ is the selling price of the watch.

According to this model,

- (a) find, using calculus, the maximum possible annual profit. (6)
- (b) Justify, also using calculus, that the profit you have found is a maximum. (2)

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7. Kim starts working for a company.

- In year 1 her annual salary will be £16 200
- In year 10 her annual salary is predicted to be £31 500

Model *A* assumes that her annual salary will increase by the same amount each year.

(a) According to model *A*, determine Kim's annual salary in year 2.

(3)

Model *B* assumes that her annual salary will increase by the same percentage each year.

(b) According to model *B*, determine Kim's annual salary in year 2. Give your answer to the nearest £10

(3)

(c) Calculate, according to the two models, the difference between the total amounts that Kim is predicted to earn from year 1 to year 10 inclusive. Give your answer to the nearest £10

(3)

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- $$8^{2x+1} = 6$$

(ii) Using the laws of logarithms, solve

$$\log_5(7-2y) = 2\log_5(y+1) - 1 \quad (5)$$

10.

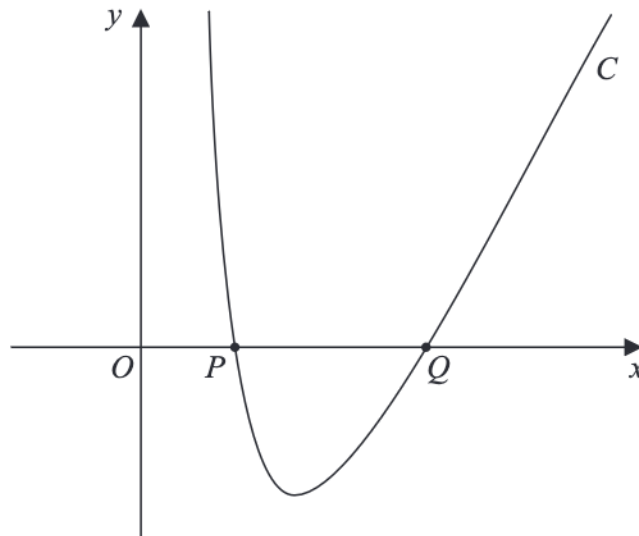


Figure 1

Figure 1 shows a sketch of part of the curve C with equation $y = f(x)$ where

$$f(x) = \frac{36}{x^2} + 2x - 13 \quad x > 0$$

Using calculus,

- (a) find the range of values of x for which $f(x)$ is increasing,

(4)

- (b) show that $\int_2^9 \left(\frac{36}{x^2} + 2x - 13 \right) dx = 0$

(4)

The point $P(2, 0)$ and the point $Q(6, 0)$ lie on C .

Given $\int_2^6 \left(\frac{36}{x^2} + 2x - 13 \right) dx = -8$

- (c) (i) state the value of $\int_6^9 \left(\frac{36}{x^2} + 2x - 13 \right) dx$

- (ii) find the value of the constant k such that $\int_2^6 \left(\frac{36}{x^2} + 2x + k \right) dx = 0$

(3)

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